

HJM model density of the shifted lognormal case with stochastic volatility

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1 Model formulation

Starting from the multi-factor HJM term structure model for the dynamics of the forward rates, driven by N correlated Brownian motions, we have

$$df(t, T) = \left(\sum_{i,j} \sigma_i(t, T) \rho_{ij}(t) \int_t^T \sigma_j(t, u) du \right) dt + \sum_i \sigma_i(t, T) dW_i(t). \quad (1)$$

Under the separability condition

$$\sigma_i(t, u) = \sigma_i(t, X) \theta_i(u) \quad (2)$$

we obtain

$$\begin{aligned} df(t, T) &= \sum_{i,j} \theta_i(T) \sigma_i(t, X) \rho_{ij}(t) \sigma_j(t, X) \left(\int_t^T \theta_j(u) du \right) dt + \sum_i \theta_i(T) \sigma_i(t, X) dW_i(t) \\ &= \sum_{i,j} \theta_i(T) d\Psi_{i,j}(t, X) \Theta_j(T) + \sum_i \theta_i(T) dX_i(t), \end{aligned}$$

where

$$\Theta_j(T) = \int_{t_0}^T \theta_j(u) du, \quad (3)$$

$$d\Psi_{i,j}(t, X) = \sigma_i(t, X) \rho_{ij}(t) \sigma_j(t, X) dt, \quad (4)$$

$$dX_i = \sigma_i(t, X) \left(dW_i(t) - \sum_j \rho_{ij}(t) \sigma_j(t, X) \Theta_j(t) dt \right), \quad (5)$$

with $X_i(t_0) = 0$ and $\Psi_{i,j}(t_0) = 0$.

Integrating then yields

$$f(t, T) = f(t_0, T) + \sum_i \theta_i(T) \left(X_i(t) + \sum_j \Psi_{i,j}(t, X) \Theta_j(T) \right),$$

$$\begin{aligned} r_t &= f(t_0, t) + \sum_i \theta_i(t) \left(X_i(t) + \sum_j \Psi_{i,j}(t, X) \Theta_j(t) \right) \\ &= f(t_0, t) + \bar{r}_t, \end{aligned}$$

$$P(t, T) = \exp \left(- \int_t^T f(t_0, u) du \right),$$

$$B(t, T) = P(t, T) Z_t \exp \left(- \sum_i \theta_i(T) \left(X_i(t) + \frac{1}{2} \sum_j \Psi_{i,j}(t, X) \Theta_j(T) \right) \right),$$

where

$$Z_t = \exp\left(\sum_i \Theta_i(t) \left(X_i(t) + \frac{1}{2} \sum_j \Psi_{i,j}(t, X) \Theta_j(t)\right)\right).$$

Introduce the Radon–Nikodym process M_t by

$$\frac{dM_t}{M_t} = \sum_i \Theta_i(t) \sigma_i(t) dW_i(t). \quad (6)$$

Theorem 1.1. *The Radon–Nikodym derivative M_t satisfies*

$$\frac{M_t}{M_s} = \exp\left(-\int_s^t \bar{r}_u du\right) \frac{Z_t}{Z_s}.$$

Under this measure the dynamics become

$$\begin{aligned} \Theta_j(T) &= \int_{t_0}^T \theta_j(u) du, \\ d\Psi_{i,j}(t, X) &= \sigma_i(t, X) \rho_{ij}(t) \sigma_j(t, X) dt, \\ dX_i &= \sigma_i(t, X) dW_i^M(t), \end{aligned}$$

and

$$E_t^Q \left[\exp\left(-\int_t^T r_u du\right) \Phi(X_T, \Psi_T) \right] = P(t, T) Z_t E_t^M \left[\frac{\Phi(X_T, \Psi_T)}{Z_T} \right].$$

Proof. Itô's formula gives

$$\Theta_i(t) X_i(t) - \Theta_i(s) X_i(s) = \int_s^t \theta_i(u) X_i(u) du + \int_s^t \Theta_i(u) dX_i(u).$$

Hence

$$\begin{aligned} \int_s^t \bar{r}_u du &= \sum_i \left(\Theta_i(t) X_i(t) - \Theta_i(s) X_i(s) - \int_s^t \Theta_i(u) \sigma_i(u) dW_i(u) \right) \\ &\quad + \sum_{i,j} \int_s^t \theta_i(u) \Psi_{i,j}(u) \Theta_j(u) du + \sum_{i,j} \int_s^t \Theta_i(u) \sigma_i(u) \rho_{ij}(u) \sigma_j(u) \Theta_j(u) du. \end{aligned}$$

Integration by parts yields

$$\begin{aligned} \sum_{i,j} \int_s^t \theta_i(u) \Psi_{i,j}(u) \Theta_j(u) du &= \frac{1}{2} \sum_{i,j} \left(\Theta_i(t) \Psi_{i,j}(t, X) \Theta_j(t) - \Theta_i(s) \Psi_{i,j}(s, X) \Theta_j(s) \right) \\ &\quad - \frac{1}{2} \sum_{i,j} \int_s^t \Theta_i(u) \sigma_i(u) \rho_{ij}(u) \sigma_j(u) \Theta_j(u) du. \end{aligned}$$

Therefore

$$\begin{aligned} \int_s^t \bar{r}_u du &= \sum_i \Theta_i(t) \left(X_i(t) + \frac{1}{2} \sum_j \Psi_{i,j}(t, X) \Theta_j(t) \right) \\ &\quad - \sum_i \Theta_i(s) \left(X_i(s) + \frac{1}{2} \sum_j \Psi_{i,j}(s, X) \Theta_j(s) \right) \\ &\quad - \sum_i \int_s^t \Theta_i(u) \sigma_i(u) dW_i(u) + \frac{1}{2} \sum_{i,j} \int_s^t \Theta_i(u) \sigma_i(u) \rho_{ij}(u) \sigma_j(u) \Theta_j(u) du. \end{aligned}$$

The first two lines are $\log Z_t - \log Z_s$. The remaining stochastic exponential is exactly M_t/M_s , which proves

$$\frac{M_t}{M_s} = \exp\left(-\int_s^t \bar{r}_u du\right) \frac{Z_t}{Z_s}.$$

Under M , the finite-variation drift in dX_i is removed by Girsanov, and the pricing identity follows from $dQ|_{\mathcal{F}_T}/dM|_{\mathcal{F}_T} = M_T^{-1}$ together with $P(t, T) = \exp(-\int_t^T f(t_0, u) du)$. $\square$

2 Shifted lognormal case with lognormal stochastic volatility

In this section we consider the one-factor case with volatility

$$\sigma(t, X) = \sigma(t) \eta_t \left(\beta(X + \gamma \Psi(t, X)) + 1 \right), \quad (7)$$

where η_t follows an uncorrelated Black–Karasinski dynamics [2]

$$d \log \eta_t = \left(m(t) - \kappa \log \eta_t \right) dt + \nu dZ_t, \quad d\langle Z, W \rangle = 0. \quad (8)$$

Set

$$\begin{aligned} S_t &= \beta(X + \gamma \Psi(t, X)) + 1, \\ A_t &= \beta^2 \Psi(t, X). \end{aligned}$$

The initial conditions $X(t_0) = \Psi(t_0) = 0$ give $S_{t_0} = 1$ and $A_{t_0} = 0$. The zero correlation is used so that the time-changed Brownian motion is independent of the clock and the Matsumoto–Yor formulae below apply. It is not a material restriction on the smile. Instantaneous volatility is $V_t = \eta_t S_t$, so dX and dV remain correlated through the local factor S even though η is independent of W :

$$\begin{aligned} dX_t &= \sigma(t) V_t dW_t^M, \\ dV_t &= \beta \sigma(t) \eta_t V_t dW_t^M + \nu \eta_t S_t dZ_t + \dots, \\ \rho_t^{\text{eff}} &= \frac{\langle dX, dV \rangle}{\sqrt{\langle dX \rangle \langle dV \rangle}} = \frac{\beta \sigma(t) \eta_t}{\sqrt{\beta^2 \sigma^2(t) \eta_t^2 + \nu^2}}. \end{aligned}$$

Thus β is the parameter that captures rate–vol correlation (the SABR β –versus– ρ trade-off). Once β is free, a correlated driver for η is weakly identified and would only add unspanned skew at the cost of destroying the time change. The remaining parameter γ is not a correlation: it mixes the quadratic-variation state Ψ into S and produces the Girsanov kernel $(\gamma/\beta)S$ below.

Using Theorem 1.1 under the M -measure we have

$$\frac{dS_t}{S_t} = \beta \sigma(t) \eta_t dW_t^M + \beta \gamma \sigma^2(t) \eta_t^2 S_t dt, \quad (9)$$

$$dA_t = \beta^2 \sigma^2(t) \eta_t^2 S_t^2 dt. \quad (10)$$

Define the stochastic time change

$$d\tau_t = \beta^2 \sigma^2(t) \eta_t^2 dt,$$

so that

$$\tau_t = \beta^2 \int_{t_0}^t \sigma^2(u) \eta_u^2 du. \quad (11)$$

Write $Y_t = \log \eta_t$. Then Y is Gaussian,

$$\begin{aligned} Y_t &= e^{-\kappa(t-t_0)} Y_{t_0} + \int_{t_0}^t e^{-\kappa(t-s)} m(s) ds + \nu \int_{t_0}^t e^{-\kappa(t-s)} dZ_s, \\ \text{Cov}(Y_t, Y_s) &= \frac{\nu^2}{2\kappa} \left(e^{-\kappa|t-s|} - e^{-\kappa(t+s-2t_0)} \right), \end{aligned}$$

and τ is the exponential functional $\tau_t = \beta^2 \int_{t_0}^t \sigma^2(u) e^{2Y_u} du$. There is no Hartman–Watson density for this integral when $\kappa \neq 0$. The Laplace transform is given by Feynman–Kac: for $\lambda > 0$,

$$\psi(t, y; \lambda) = E \left[\exp \left(-\lambda \beta^2 \int_t^T \sigma^2(u) e^{2Y_u} du \right) \middle| Y_t = y \right]$$

solves

$$\begin{aligned} \partial_t \psi + (m(t) - \kappa y) \partial_y \psi + \frac{\nu^2}{2} \partial_{yy} \psi - \lambda \beta^2 \sigma^2(t) e^{2y} \psi &= 0, \\ \psi(T, y; \lambda) &= 1, \end{aligned}$$

so $E[e^{-\lambda \tau_T} | Y_{t_0} = y_0] = \psi(t_0, y_0; \lambda)$. The density follows by Bromwich inversion,

$$p(\tau_T \in du) = \frac{du}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{\lambda u} \psi(t_0, y_0; \lambda) d\lambda.$$

If $\kappa = 0$ and m, σ are constant, Y is Brownian motion with drift and τ_T reduces to a scaled Yor functional $A^{(\mu)}$; its density is then Hartman–Watson [1]. In the general BK case the moments of τ are nevertheless closed, because (Y_u) is Gaussian:

$$\begin{aligned} E[\eta_u^2] &= \exp(2E[Y_u] + 2\text{Var}(Y_u)), \\ E[\tau_T] &= \beta^2 \int_{t_0}^T \sigma^2(u) E[\eta_u^2] du, \\ \text{Var}(\tau_T) &= \beta^4 \int_{t_0}^T \int_{t_0}^T \sigma^2(u) \sigma^2(v) E[\eta_u^2] E[\eta_v^2] (e^{4\text{Cov}(Y_u, Y_v)} - 1) du dv. \end{aligned}$$

Matching these two moments to a lognormal law of τ_T gives a usable mixing density; the exact law is the Laplace inversion above.

Then, under M and in the time-changed clock,

$$\frac{dS_\tau}{S_\tau} = dW_\tau^M + \frac{\gamma}{\beta} S_\tau d\tau, \quad (12)$$

$$dA_\tau = S_\tau^2 d\tau. \quad (13)$$

Girsanov with kernel $(\gamma/\beta)S$ removes the finite-variation drift. The resulting density is

$$L_\tau := \frac{dQ^D}{dQ^M} = \exp \left(-\frac{\gamma}{\beta} (S_\tau - 1) + \frac{\gamma^2}{2\beta^2} A_\tau \right),$$

hence

$$\frac{dQ^M}{dQ^D} = L_\tau^{-1} = \exp\left(\frac{\gamma}{\beta}(S_\tau - 1) - \frac{\gamma^2}{2\beta^2}A_\tau\right).$$

Under Q^D the dynamics become

$$\begin{aligned} S_\tau &= \exp\left(W_\tau^D - \frac{\tau}{2}\right), \\ A_\tau &= \int_0^\tau S_u^2 du, \\ U_\tau &= \frac{S_\tau}{A_\tau}. \end{aligned}$$

In particular $S_\tau = \exp(B_\tau^{(-1/2)})$ and $A_\tau = A_\tau^{(-1/2)}$ in the notation of Matsumoto and Yor [1]. Pricing identities under M are rewritten under D by

$$E^M\left[\frac{g}{Z}\right] = E^D\left[L^{-1}\frac{g}{Z}\right].$$

The true one-factor HJM bond is, with $X = (S - 1)/\beta - \gamma A/\beta^2$ and $\Psi = A/\beta^2$,

$$\frac{B(t, T)}{Z_t} = P(t, T) \exp\left(-\frac{\Theta(T)}{\beta}(S_\tau - 1) + \frac{\gamma\Theta(T)}{\beta^2}A_\tau - \frac{\Theta(T)^2}{2\beta^2}A_\tau\right).$$

Multiplying by the Radon–Nikodym factor L^{-1} produces the D -measure kernel

$$L_\tau^{-1}\frac{B(t, T)}{Z_t} = P(t, T) \exp\left(-\left(\frac{\Theta(T) - \gamma}{\beta}\right)(S_\tau - 1) - \frac{A_\tau}{2}\left(\frac{\Theta(T) - \gamma}{\beta}\right)^2\right).$$

(The previous display is *not* the bond B/Z ; it is the integrand for an expectation under Q^D .)

Write $\alpha(T) = (\Theta(T) - \gamma)/\beta$. The joint law of (S_τ, U_τ) under Q^D , conditionally on the clock τ (i.e. on the path of η), follows from [1, Thm. 4.1] after the change of variables $U = S/A$. The conditional law of S given $U = u$ is inverse Gaussian with mean 1 and shape u ,

$$p(S_\tau = x \mid U_\tau = u, \tau) = \sqrt{\frac{u}{2\pi x^3}} \exp\left(u - \frac{u}{2}\left(x + \frac{1}{x}\right)\right),$$

and the marginal of U is

$$p(U_\tau = u \mid \tau) = \sqrt{2\pi} \exp\left(-u - \frac{\tau}{8}\right) \frac{H(u, \tau)}{u^{3/2}},$$

where H is the Hartman–Watson function θ of [1]. The conditional density in S does not depend on τ .

The product of this density with the D -kernel, divided by $P(t, T)$, is the density of an inverse Gaussian law with mean $1/\mu$ and shape U_τ , where $\mu = 1 + \alpha(T)/U_\tau = 1 + (\Theta(T) - \gamma)/(\beta U_\tau)$:

$$\begin{aligned}
p(x \mid U_\tau, \tau) L^{-1} \frac{B(t, T)}{P(t, T) Z_t} \Big|_{S=x} &= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(\alpha(T) + U_\tau - \frac{1}{2} \left(\frac{x}{U_\tau} (U_\tau + \alpha(T))^2 + \frac{U_\tau}{x} \right)\right) \\
&= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(-\frac{U_\tau}{2x} \left(x\mu - 1\right)^2\right) \\
&= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(-\frac{1}{2} \left(\sqrt{U_\tau x} \mu - \sqrt{\frac{U_\tau}{x}}\right)^2\right) \\
&= \sqrt{\frac{U_\tau}{2\pi x^3}} \exp\left(2\mu U_\tau - \frac{1}{2} \left(\sqrt{U_\tau x} \mu + \sqrt{\frac{U_\tau}{x}}\right)^2\right).
\end{aligned}$$

(The four right-hand sides are identical; they are a density times a kernel, not the bond itself.)

Define

$$\begin{aligned}
\xi^+(k) &= \sqrt{U_\tau k} \mu - \sqrt{\frac{U_\tau}{k}}, \\
\xi^-(k) &= -\sqrt{U_\tau k} \mu - \sqrt{\frac{U_\tau}{k}}.
\end{aligned}$$

At frozen U_τ one has $d\xi^+ + d\xi^- = \sqrt{U_\tau/k^3} dk$, but the two branches carry different Jacobians; the Gaussian split uses each branch separately. Integrating the product against $\mathbf{1}_{\{S \leq k\}}$ yields the inverse-Gaussian cdf (mean $1/\mu$, shape U_τ)

$$E^D \left[L^{-1} \frac{B(t, T)}{Z_t} \mathbf{1}_{\{S_\tau \leq k\}} \mid U_\tau, \tau \right] = P(t, T) \left(N(\xi^+(k)) + \exp\left(2 \left(U_\tau + \frac{\Theta(T) - \gamma}{\beta} \right)\right) N(\xi^-(k)) \right).$$

A payer swaption with expiry T_e , last payment T_n and strike K has time- T_e value $(1 - B(T_e, T_n) - K \sum_{i=e+1}^n \delta_i B(T_e, T_i))^+$, using $B(T_e, T_e) = 1$. Theorem 1.1 gives

$$\begin{aligned}
\text{swap}(t) &= P(t, T_e) Z_t E^M \left[\frac{\left(1 - B(T_e, T_n) - K \sum_{i=e+1}^n \delta_i B(T_e, T_i)\right)^+}{Z_{T_e}} \right] \\
&= P(t, T_e) Z_t E^D \left[L^{-1} \left(G_e - G_n - K \sum_{i=e+1}^n \delta_i G_i \right)^+ \right],
\end{aligned}$$

where $G_i := B(T_e, T_i)/Z_{T_e}$. Given U , every G_i is strictly monotone in S , so the forward swap value has a unique root S^* and the positive part is the same linear combination restricted to $\{S > S^*\}$ (or $\{S < S^*\}$, according to the sign of β). Each term $E^D[L^{-1} G_i \mathbf{1}_{\{S \geq S^*\}} \mid U, \tau]$ is the inverse-Gaussian formula above with tenor-specific $\alpha(T_i) = (\Theta(T_i) - \gamma)/\beta$. The remaining average is over the Hartman–Watson law of U given τ and over the law of the clock τ obtained above (Laplace inversion of ψ , or the lognormal mix fitted to $E[\tau]$ and $\text{Var}(\tau)$).

3 One-factor Cheyette exotics

The Section 2 mix is the law of the curve at a *single* date. It calibrates $(\sigma(t), \beta, \gamma, \kappa, m, \nu)$ to European swaptions. Bermudans and lookbacks see the curve along a schedule, so they are priced as one-factor Cheyette Markovian claims on (X, Ψ) , with η an extra state only because volatility is stochastic [3]. Spreads, basis and FX-linked coupons are outside this setup.

Under Q ,

$$dX_t = \sigma(t) \eta_t S_t \left(dW_t - \sigma(t) \eta_t S_t \Theta(t) dt \right), \quad (14)$$

$$d\Psi_t = \sigma^2(t) \eta_t^2 S_t^2 dt, \quad (15)$$

$$d \log \eta_t = (m(t) - \kappa \log \eta_t) dt + \nu dZ_t, \quad d\langle Z, W \rangle = 0, \quad (16)$$

with $S_t = \beta(X_t + \gamma \Psi_t) + 1$. The whole curve is a closed function of (X, Ψ) :

$$\begin{aligned} r_t &= f(t_0, t) + \theta(t) (X_t + \Psi_t \Theta(t)), \\ B(t, T) &= P(t, T) Z_t \exp \left(-\Theta(T) \left(X_t + \frac{1}{2} \Psi_t \Theta(T) \right) \right), \\ L(t; T, T + \delta) &= \frac{1}{\delta} \left(\frac{B(t, T)}{B(t, T + \delta)} - 1 \right), \\ R(t; T_a, T_b) &= \frac{B(t, T_a) - B(t, T_b)}{\sum \delta_i B(t, T_i)}. \end{aligned}$$

Given Ψ , every such rate is strictly monotone in X . That is the one-factor Cheyette property used below. Europeans, Bermudans and lookbacks are all valued with the same generalised inverse-Gaussian (GIG) transition, conditioned on the current state and on the clock increment.

Conditional GIG transition. On $[T_i, T_{i+1}]$ write $\Delta\tau_i = \beta^2 \int_{T_i}^{T_{i+1}} \sigma^2(u) \eta_u^2 du$ and let $(\hat{S}, \hat{A})$ be the Matsumoto–Yor pair started from 1, run for time $\Delta\tau_i$ under Q^D . The Cheyette factor maps as

$$S' = S \hat{S}, \quad A' = A + S^2 \hat{A}, \quad \hat{U} = \hat{S} / \hat{A}.$$

Conditionally on $\hat{U} = u$, $\hat{S}$ is inverse Gaussian with mean 1 and shape u , i.e. $\text{GIG}(-\frac{1}{2}, u, u)$. Scaling gives the transition in S :

$$\begin{aligned} S' \mid S, \hat{U} = u &\sim \text{GIG} \left(-\frac{1}{2}, uS, \frac{u}{S} \right), \\ p(s' \mid s, u) &= \sqrt{\frac{us}{2\pi s'^3}} \exp \left(u - \frac{u}{2} \left(\frac{s'}{s} + \frac{s}{s'} \right) \right), \\ A' &= A + \frac{SS'}{u}. \end{aligned}$$

Given (S, u) , A' is deterministic: the two-dimensional kernel in (S, A) collapses to a one-dimensional GIG in S' . On a grid this is the GIG transition matrix $G(s, s'; u)$. The

increment $\widehat{U}$ is mixed with the Hartman–Watson law at time $\Delta\tau_i$, and $\Delta\tau_i$ is mixed with the Black–Karasinski law of the clock conditional on η_{T_i} (Section 2). The incremental Girsanov is

$$L_{i \rightarrow i+1} = \exp\left(-\frac{\gamma}{\beta}(S' - S) + \frac{\gamma^2}{2\beta^2}(A' - A)\right).$$

A cashflow Φ paid at T_{i+1} therefore has value

$$P(T_i, T_{i+1}) Z_{T_i} E^D \left[L_{i \rightarrow i+1}^{-1} \frac{\Phi}{Z_{T_{i+1}}} \middle| S, A, \eta_{T_i} \right],$$

the inner expectation being the GIG matrix, the outer the mix over $(\widehat{U}, \Delta\tau_i, \eta_{T_{i+1}}) \mid \eta_{T_i}$. A European swaption is the same kernel in one step from t_0 ($S = 1, A = 0$).

Bermudan swaption. Fix exercise dates $T_1 < \dots < T_N$ and strike K . At T_i the exercise value is the remaining vanilla swap, hence a function of the current curve only:

$$E_i(S, A) = \left(B(T_i, T_i) - B(T_i, T_n) - K \sum_{j=i+1}^n \delta_j B(T_i, T_j) \right)^+.$$

No nested option is hidden in E_i . Let $u_i(S, A, \eta)$ be the value after the exercise decision. Then $u_N = E_N$ and $u_i = \max(E_i, C_i)$, where C_i is one application of the conditional GIG matrix to u_{i+1} :

$$C_i(S, A, \eta) = P(T_i, T_{i+1}) Z_i E \left[G(S, S'; \widehat{U}) L_{i \rightarrow i+1}^{-1} \frac{u_{i+1}(S', A', \eta')}{Z'} \middle| \eta \right].$$

The GIG matrix is always used. The only change relative to a European is that the kernel is applied *conditionally* on the state at T_i and chained backwards. One must not mix ready-made one-factor Bermudans over the unconditional law of τ_{T_N} : that would ignore the filtration at each T_i . Conditioning on $(\widehat{U}, \Delta\tau_i, \eta_{T_i})$ restores the Markov property and keeps the same kernel as Section 2.

Lookback. Let $R_t = R(t, S_t, A_t)$ be a monotone Cheyette rate and $\underline{R}_t = \min_{u \leq t} R_u$ (resp. $\overline{R}_t$ the max). A fixed-strike lookback pays $(K - \underline{R}_T)^+$ or $(\overline{R}_T - K)^+$; a floating-strike lookback pays $(R_T - \underline{R}_T)^+$ or $(\overline{R}_T - R_T)^+$. Discrete fixings $t_1, \dots, t_n$: between t_k and t_{k+1} the running extremum is frozen, so the continuation is again the conditional GIG matrix in (S, A) given η_{t_k} . At t_{k+1} update $\underline{R} \leftarrow \min(\underline{R}, R(t_{k+1}, S', A'))$ and continue. The European mix is not used; the GIG transition is, at every step, conditional on the recorded extremum and on η .

References

- [1] H. Matsumoto and M. Yor, Exponential functionals of Brownian motion, I: Probability laws at fixed time, *Probability Surveys* **2** (2005), 312–347.
- [2] F. Black and P. Karasinski, Bond and option pricing when short rates are lognormal, *Financial Analysts Journal* **47** (1991), 52–59.
- [3] L. B. G. Andersen and V. V. Piterbarg, *Interest Rate Modeling. Volume I: Foundations and Vanilla Models*, Atlantic Financial Press, 2010.